

CHALLENGE – 9 – CRYPTOGRAPHY

substitution2 

Medium Cryptography picoCTF 2022 Substitution

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Description

It seems that another encrypted message has been intercepted. The encryptor seems to have learned their lesson though and now there isn't any punctuation! Can you still crack the cipher?

Download the message [here](#).

Hints ?

1

Try refining your frequency attack, maybe analyzing groups of letters would improve your results?

19,068 users solved

86% Liked

picoCTF{FLAG}

Submit Flag

Description

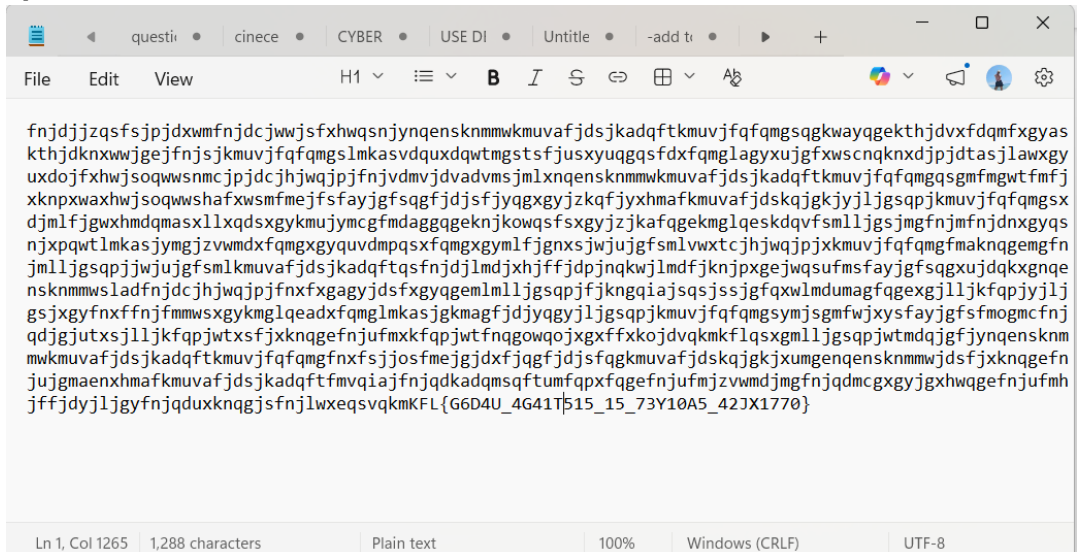
The challenge involves an intercepted encrypted message where the encryptor has removed all punctuation to make cracking the cipher more difficult. The goal is to decrypt the provided text file and retrieve the flag.

Solution Approach

1. Analyze the Ciphertext

The provided ciphertext is a dense block of lowercase letters without

s

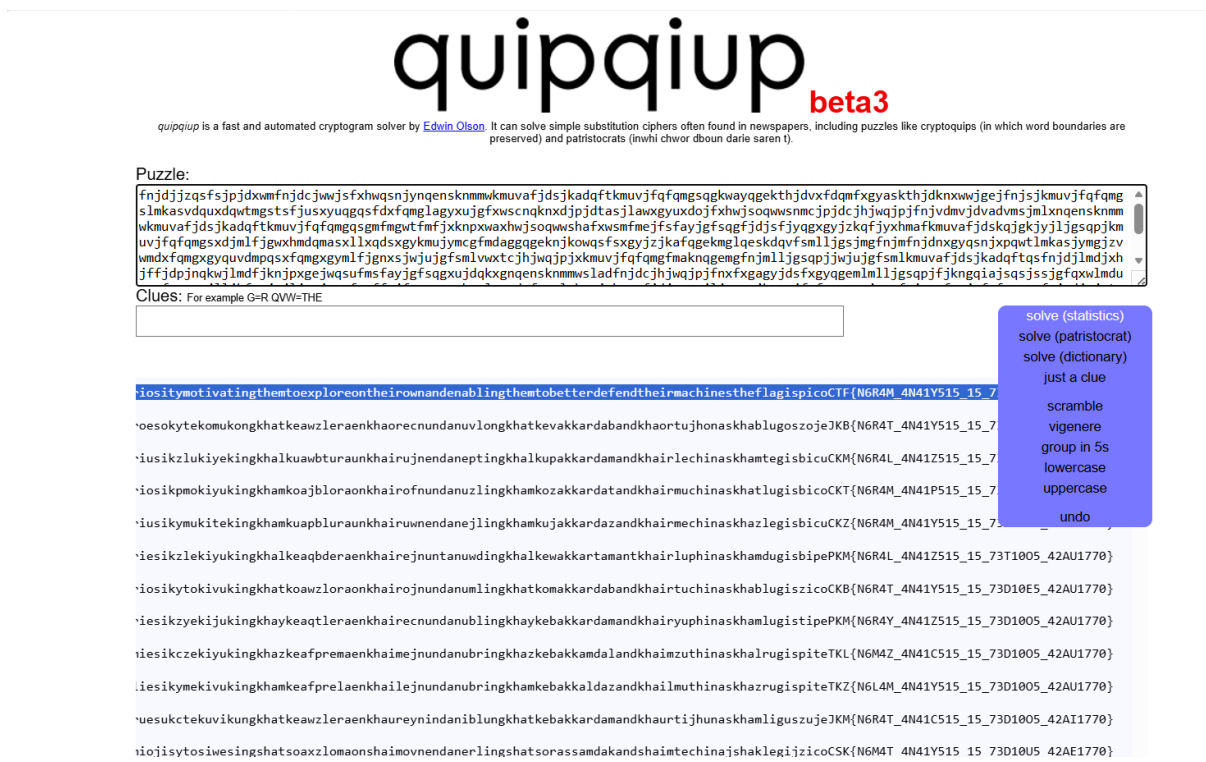


```
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jffjdyjljgyfnjqduknqgjsfnjlwxeqsvqkmKFL{G6D4U_4G41Tf515_15_73Y10A5_42JX1770}
```

2. Automated Decryption (quipqiup)

Due to the lack of word boundaries (a "patristocrat" style cipher), an automated tool like **quipqiup** was used.

- The tool analyzes the frequency of letter groupings to map the ciphertext to the English alphabet.
- By providing the ciphertext to the solver, multiple potential plaintexts were generated.



3. Plaintext Recovery

The decryption reveals a text discussing the educational value of high school computer security competitions. Key decoded terms include:

- kmuvjfqfmg → **competition**.
- fmaknqge → **teaching**.
- fnj → **the**.

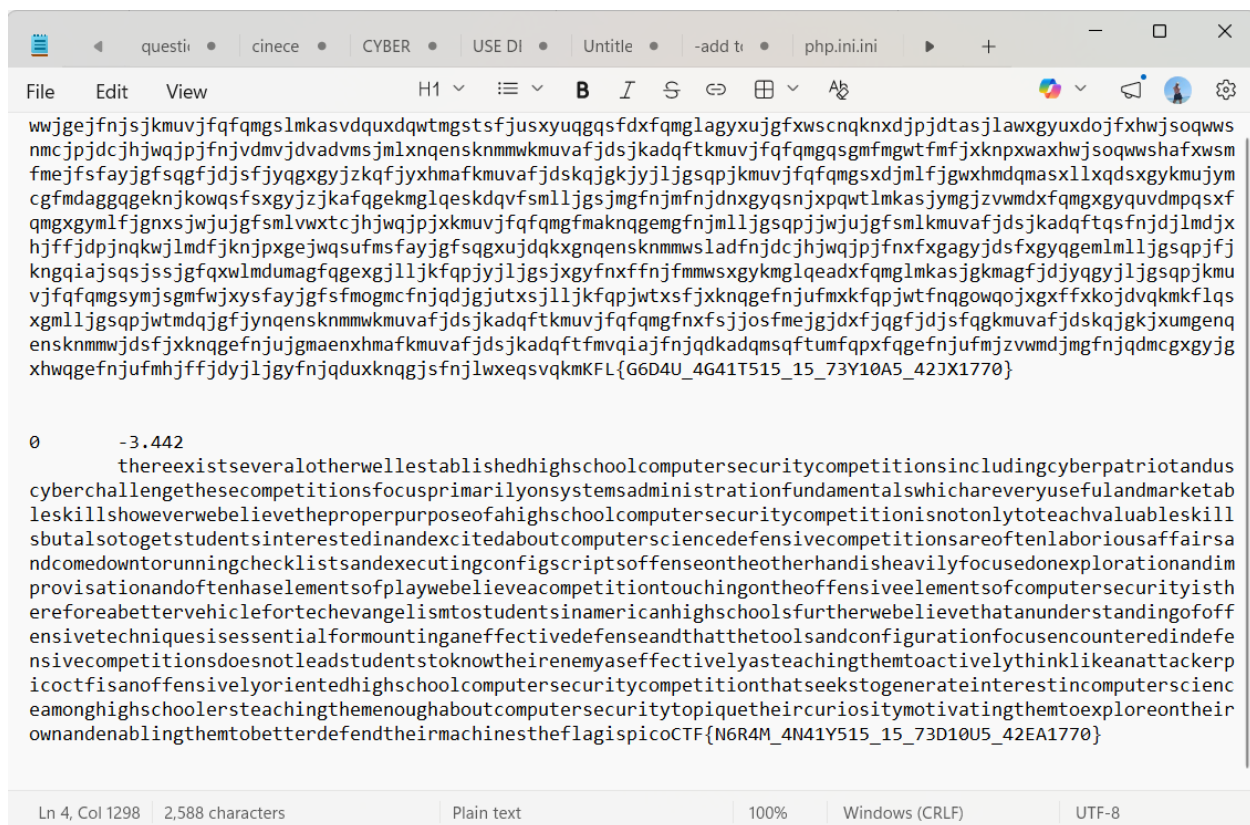
4. Retrieval of the Flag

The flag is located at the very end of the decrypted message. The decrypted text concludes with:

"...enabling them to better defend their machines the flag is picoCTF{N6R4M_4N41Y515_15_7301005_42EA1770}".

Final Flag

picoCTF{N6R4M_4N41Y515_15_7301005_42EA1770}



```
wwjgejfnjsjkmuvjfqqfmgslmkasvdquxdqwtmgstsfjusxyuqggsfdxfqmglagyxujgfwscnqkxndjppdtasjlawxyuxdojfxhwjsoqwws
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0 -3.442

there exist several other well established high school computer security competitions including cyberpatriot and us cyber challenge these competitions focus primarily on systems administration fundamentals which are very useful and marketable skills however we believe the proper purpose of a high school computer security competition is not only to teach a valuable skill but also to get students interested in and excited about computer science defensive competitions are often laborious affairs and come down to running checklists and executing config scripts offense on the other hand is heavily focused on exploration and improvisation and often has elements of play we believe a competition touching on the offensive elements of computer security is therefore a better vehicle for tech evangelism to students in American high schools further we believe that an understanding of offensive techniques is essential for mounting an effective defense and that the tools and configuration focus encountered in defensive competitions does not lead students to know their enemy as effectively as teaching them to actively think like an attacker picoctf is an offensively oriented high school computer security competition that seeks to generate interest in computer science among high schoolers teaching them enough about computer security to pique their curiosity motivating them to explore on their own and enabling them to better defend their machines the flag is picoCTF{N6R4M_4N41Y515_15_73D10U5_42EA1770}

Ln 4, Col 1298 2,588 characters Plain text 100% Windows (CRLF) UTF-8